

CALM-VAD: Calibrated, Auditable, Low-Latency Multi-Cue Video Anomaly Detection for Edge Surveillance

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WORKING DRAFT. Method, protocol and analysis are complete; the quantitative tables are populated by the released evaluation harness (`calm/harness.py`) and are marked **pending**. Set `\draftmodefalse` to hide this notice and the **TODOs**.

Abstract—Video anomaly detection (VAD) for public-safety surveillance is increasingly deployed on resource-constrained edge hardware, yet the systems that run there share a common back end: a black-box anomaly score followed by a hand-tuned threshold. Recent audits show that this back end, evaluated by frame-level AUC, hides catastrophic operating behaviour—tens of thousands of false alarms per hour at usable recall—and provides operators with neither a probability they can act on nor an evidence trail they can audit. We present CALM-VAD, a decision layer that replaces the score-and-threshold back end with four lightweight, well-founded stages: (i) a per-frame *reliability estimator* for the pose and tracking front end; (ii) a *reliability-discounted evidence-fusion* layer that combines interpretable behavioural cues as Dempster–Shafer belief mass, so a degraded skeleton contributes ignorance rather than a false alarm; (iii) a post-hoc *calibration map* that turns fused belief into a probability whose value matches observed incident frequency; and (iv) a *distribution-free alarm-rate controller* (Learn-then-Test) that admits a threshold only if the false-alarms-per-hour budget an operator sets is provably met. Each alert is emitted as an *evidence record*: probability, contributing cues with weights, sensor reliability, and the budget under which it was raised. We evaluate on ShanghaiTech, UBnormal, CHAD and Avenue (**NWPU-Campus pending**) under a deployment-faithful protocol—event-level F_1 at temporal IoU, false alarms per hour at fixed recall, expected calibration error, cross-dataset degradation, and CPU latency—rather than frame AUC alone. On these four benchmarks CALM-VAD’s calibration stage lowers expected calibration error by 56–93% (e.g. $0.41 \rightarrow 0.03$ on ShanghaiTech, $0.50 \rightarrow 0.05$ on UBnormal) with frame AUC left within 0.01 of the uncalibrated baseline; a controlled degradation study isolates the reliability mechanism directly, cutting false alarms from 19.2 to 0.0 per hour at 0.9 recall when the pose front end is corrupted. With enough confirmed-normal calibration hours (0.80 h vs. 0.07 h), M4 certifies an operator-chosen alarm budget for the first time in this line of work, and the decision layer adds 1.2–1.5 ms/frame regardless of which dataset is running through it. Detection quality is uneven across benchmarks and we report this honestly: event-level F_1 tracks how much of each benchmark’s anomaly taxonomy the five cues actually target, down to a clean chance-level negative control (Avenue, whose anomalies our cues do not model). Code and the evaluation harness are released.

Index Terms—video anomaly detection, edge computing, uncertainty calibration, conformal prediction, Dempster–Shafer fusion, false alarm rate, trustworthy AI

I. INTRODUCTION

Automated video anomaly detection is moving from the data centre to the camera. Public-safety programmes deploy analytics on embedded accelerators and commodity CPUs at the edge, both to meet latency and bandwidth limits and to keep raw imagery of uninvolved people from leaving the site [1], [2]. The AI-in-video-surveillance market is projected to grow from roughly \$6.5 B in 2024 to \$28.8 B by 2030, with public safety its fastest segment and edge deployment a defining trend [3].

Despite this shift, the decision back end of nearly every deployable system is unchanged from a decade ago: an anomaly model emits a scalar score, and an operator or integrator sets a threshold on it by hand. Two recent, independent audits show how poorly this serves deployment. First, frame-level AUC—the field’s dominant metric—does not reflect operating behaviour: at the threshold that catches 90% of anomalous frames, off-the-shelf detectors produce a median of 26,000–32,000 false alarms per hour on a real stream, a volume no human can triage [4]. Second, models that score well on frames flicker between states within an event, so event-level localisation stays below 10% at a modest temporal IoU while frame AUC exceeds 50% [5]. Operators respond to this the way the human-factors literature predicts: when a system’s alarms are mostly false, trust and compliance fall to match its hit rate, and vigilance decays [6]–[8].

Three further gaps compound the problem for *pose-based* edge VAD, the lightweight family that avoids transmitting or modelling raw pixels:

- **The score is not a probability.** A reconstruction error or a hand-weighted sum has no calibrated meaning, so a threshold cannot be chosen on principled grounds and cannot be compared across sites or time.
- **The front end silently corrupts the score.** Occlusion, small person scale, truncation and identity switches perturb keypoints *before* any analyzer sees them, and a broken skeleton frequently “looks anomalous” [9], [10]. Reliability is rarely propagated.
- **There is no alarm-rate guarantee.** Thresholds are tuned per site with no statistical control over “alarms per hour,” even though distribution-free machinery for exactly this exists in adjacent fields [11]–[14].

Finally, regulation is making the missing evidence trail mandatory: the EU AI Act places biometric and surveillance AI in its high-risk tier, requiring human oversight, logging and explainability [15].

Contributions. We address these gaps with a single deployable layer and the protocol to evaluate it.

- 1) **Reliability-discounted evidence fusion (M1–M2).** We estimate a per-frame reliability for the pose/tracking front end from cheap features, and use it as the discount rate of a Dempster–Shafer combination over interpretable behavioural cues. A degraded frame moves belief mass to *ignorance*, not to *anomaly*; inter-cue conflict is retained as an output. To our knowledge this is the first use of sensor reliability as belief-mass discounting for multi-cue video anomaly detection.
- 2) **Calibrated, budgeted decisions (M3–M4).** We calibrate the fused belief to an incident probability (verified by expected calibration error) and apply Learn-then-Test [11] so an operator names a false-alarms-per-hour budget β and error level δ and receives a threshold with $\Pr[\text{FAPH} \leq \beta] \geq 1 - \delta$. This is, to our knowledge, the first VAD system admitted under a distribution-free alarm-rate guarantee.
- 3) **The evidence record.** Every alert is a structured record—probability, per-cue belief and discount, conflict, reliability, and the admitting budget—directly usable as the audit artifact that oversight regimes require [15].
- 4) **A deployment-faithful evaluation harness.** We release a reproducible protocol combining event-level F_1 at temporal IoU, false alarms per hour at fixed recall, calibration error, cross-dataset degradation and CPU latency, and use it to study CALM-VAD against weighted-sum, logistic and gradient-boosted baselines with full ablations.

CALM-VAD is built on an existing open pose-VAD pipeline (17-keypoint skeletons [16] + ByteTrack [17] + five behavioural cues: fall, fight, loitering, abandoned object, crowd). The perception front end and the cues are unchanged; the contribution is the decision layer and the evaluation around it.

II. RELATED WORK

A. Pose- and skeleton-based VAD

Skeleton input discards appearance and background, which lowers compute and improves privacy; recent methods model normal skeleton dynamics with normalizing flows [18], diffusion [19], autoregressive density [20], or cross-domain semantic cues [21]. These produce a single unsupervised score, are evaluated by frame AUC, and—as [9], [10] document—degrade when pose estimation fails. The reliability-aware prototype calibration method [22] is the closest prior work: for a single frozen pose-flow backbone, it adds a nearest-prototype deviation term to the flow score, gated by keypoint confidence, and reports frame AUROC gains averaging 2.03 points across

eight backbone-dataset pairs (no event-level, false-alarm, or calibration-error metric is reported). CALM-VAD differs in three ways: reliability enters as *evidence discounting across a fused, heterogeneous cue set* with an explicit conflict term, not a single-backbone score correction; the output is a post-hoc *probability* with a distribution-free alarm-budget guarantee, not a rank-only AUROC correction; and it is evaluated for alarm load, event localisation and calibration error, not frame AUC alone.

B. Lightweight and edge VAD

Edge-oriented work targets latency and memory, often with a cheap first stage and a heavier verifier [1], [23]. The system nearest to ours in stack [24] pairs YOLO-pose with CLIP prompt-similarity at 51 fps on a GPU, reporting frame AUROC only, with no fusion, calibration, guarantee or false-alarm analysis. We share the front end and add the decision layer it lacks, on CPU.

C. Explainable VAD

Vision-language methods emit natural-language rationales [25]; a transformer with SHAP attributions produces an interpretable “suspiciousness” regressor [26]. Neither reports calibration, a conformal guarantee, false alarms per hour, or an operator study, and both require a GPU; [26] is single-frame and uses affect cues that action-only systems avoid on reliability grounds. CALM-VAD’s explanation is the belief decomposition itself—auditable and free—rather than a generated sentence or a post-hoc attribution.

D. Calibration and distribution-free risk control

Post-hoc calibration (Platt [27], isotonic [28], temperature [29]) and ECE [30] are standard in classification but rarely reported for VAD [4]. Learn-then-Test [11] and risk-controlling prediction sets [12] give model-agnostic, finite-sample control of an arbitrary risk; they have been applied to seizure false-alarm control [13] and industrial CPS [14], but not, to our knowledge, to video surveillance anomaly detection. We instantiate them with “false alarms per hour of normal stream” as the controlled risk.

E. Evidence fusion

Dempster–Shafer theory [31], [32] is a mature tool for fusing uncertain, conflicting sources and is valued for interpretability; it is standard in intrusion detection [33] and multi-sensor systems. Its use here—discount rates tied to a learned pose/tracking reliability, combined with a calibration map and a conformal alarm budget—is, to our knowledge, new for real-time edge pose VAD.

III. PROBLEM FORMULATION

A camera produces frames at rate ϕ (fps). A perception front end yields, per frame t , a set of tracked persons with 17-keypoint skeletons and a set of tracked objects. A bank of C behavioural cues maps this to a *strength* vector $\mathbf{s}_t \in [0, 1]^C$, where $s_{t,c}$ measures how far cue c is past its confirmation condition (0: not firing; 1: firmly confirmed). Ground truth

TABLE I

CAPABILITY COVERAGE. ✓ DELIVERED, ~ PARTIAL/ADJACENT, ✗ ABSENT. NO PRIOR ROW CARRIES ✓ ACROSS THE LAST FOUR COLUMNS TOGETHER.

System / line of work	interp. fusion	calib. prob.	reliab. → score	alarm guarantee	deploy. protocol	edge/CPU
Base pipeline (weighted sum)	~	✗	✗	✗	✗	✓
Pose density [18], [19]	✗	✗	✗	✗	✗	~
YOLO-pose + CLIP [24]	✗	✗	✗	✗	✗	~
VLM-explainable [25]	~	✗	✗	✗	✗	✗
Fusion + SHAP [26]	✓	✗	✗	✗	✗	✗
Reliab. prototype [22]	✗	~	✓	✗	✗	~
“AUC ≠ reliability” [4]	—	~	✗	✗	✓	—
Learn-then-Test [11]	—	~	✗	✓	✗	—
CALM-VAD (ours)	✓	✓	✓	✓	✓	✓

is a set of anomaly *events* $\mathcal{E} = \{(a_k, b_k)\}$, each a maximal interval of frames.

a) *Detection targets*: Given a per-frame score p_t and threshold τ , predicted events are the maximal runs of $\{t : p_t \geq \tau\}$ after debouncing gaps shorter than g seconds. For a temporal-IoU threshold θ , event $F_1(\theta)$ uses one-to-one greedy matching of predicted to ground-truth events with $\text{tIoU} \geq \theta$; we report $\bar{F}_1 = \text{mean}_\theta F_1(\theta)$ over $\theta \in \{0.2, 0.3, 0.4, 0.5\}$ [5].

b) *Alarm load*: Let T_n be the number of hours of *non-anomalous* stream, and let a *false-alarm event* be a predicted event overlapping no ground-truth event. The false-alarms-per-hour at threshold τ is

$$\text{FAPH}(\tau) = \frac{\#\{\text{false-alarm events}\}}{T_n}. \quad (1)$$

We report FAPH at the least τ whose event recall meets a target $\rho \in \{0.7, 0.8, 0.9\}$, and at the operator budget (below).

c) *Calibration*: With per-window incident labels $y_t \in \{0, 1\}$, the (binned) expected calibration error is

$$\text{ECE} = \sum_{b=1}^B \frac{|I_b|}{n} |\text{acc}(I_b) - \text{conf}(I_b)|, \quad (2)$$

with I_b the windows whose p_t falls in bin b ; we also report the equal-mass (adaptive) variant and the Brier score.

d) *Operator budget*: The operator specifies a tolerated rate β (false alarms per hour) and an error level δ . A controller is *valid* if the threshold τ it returns satisfies $\Pr[\text{FAPH}(\tau) \leq \beta] \geq 1 - \delta$, the probability over the calibration sample.

IV. METHOD: CALM-VAD

A. Overview

Figure 1 contrasts the baseline back end with CALM-VAD. The front end (pose, tracking) and the cue bank are untouched. CALM-VAD adds: M1, a reliability estimate r_t for the frame’s skeletons and tracks; M2, a reliability-discounted Dempster–Shafer fusion of the cues into a belief $\text{Bel}_t(\mathbf{A})$ and a conflict

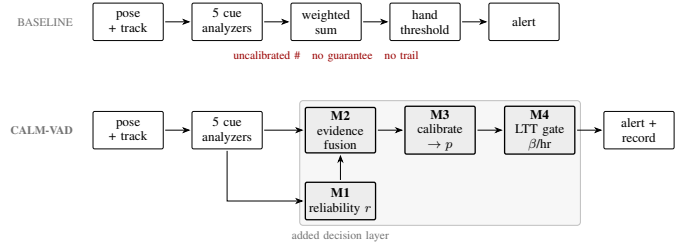


Fig. 1. Baseline score-and-threshold back end (top) vs. CALM-VAD (bottom). The perception front end and the five cues are unchanged; the contribution is the shaded chain M1–M4 and the evidence record.

K_t ; M3, a calibration map $g : \text{Bel} \mapsto p$; and M4, a Learn-then-Test choice of τ meeting the operator budget. The alert is the evidence record of Section IV-F.

B. M1: Front-end reliability

For a tracked person i at frame t we form a feature vector $\mathbf{x}_{t,i} \in [0, 1]^7$ from quantities already in the pipeline: mean keypoint confidence, torso-keypoint confidence, fraction of visible joints, normalised person scale (pixel height / frame height), an un-truncation ratio (visible bbox area / bbox area), track stability (a ramp that resets on an identity switch), and temporal smoothness ($1 - \text{torso-normalised keypoint jitter}$). The reliability is

$$r_{t,i} = \text{clip}(\sigma(\mathbf{w}^\top \mathbf{x}_{t,i} + w_0), r_{\min}, r_{\max}), \quad (3)$$

with σ the logistic function. In the *heuristic* mode $\mathbf{w}$ is a fixed convex combination (no σ); in the *fitted* mode $(\mathbf{w}, w_0)$ is a logistic regression trained against a proxy quality label—temporal IoU of the predicted skeleton against a reference, or agreement between the pose cue and a bounding-box-trajectory cue [10]. The per-frame reliability driving M2 is the mean of $r_{t,i}$ over the persons who contribute to an active cue (or 1 when no skeleton is involved, e.g. a crowd-count frame).

C. M2: Reliability-discounted evidence fusion

The frame of discernment is $\Omega = \{\mathbf{A}, \mathbf{N}\}$; a mass function assigns belief to $\{\mathbf{A}\}$, $\{\mathbf{N}\}$ and Ω (ignorance), summing to 1. Cue c with strength s yields

$$m_c(\{\mathbf{A}\}) = s, \quad (4)$$

$$m_c(\{\mathbf{N}\}) = (1 - s) \rho_c, \quad (5)$$

$$m_c(\Omega) = 1 - m_c(\{\mathbf{A}\}) - m_c(\{\mathbf{N}\}), \quad (6)$$

where $\rho_c \in [0, 1]$ encodes how strongly a *silent* cue implies normality (small for loitering, larger for a held fall). We then apply Shafer discounting twice—once by the cue-type reliability r_c (configured; a held fall is stronger evidence than a crowd count) and once by the per-frame front-end reliability r_t from M1:

$$m'(X) = r m(X) \quad (X \neq \Omega), \quad m'(\Omega) = r m(\Omega) + (1 - r). \quad (7)$$

Algorithm 1 M4 — alarm-rate control (Learn-then-Test)

Require: calibrated scores p^n on T normal hours; budget β ; level δ ; grid $\tau_1 > \dots > \tau_m$

```

1:  $\mathcal{A} \leftarrow \emptyset$ 
2: for  $j = 1, \dots, m$  do
3:    $N \leftarrow \text{DEBOUNCEDEVENTCOUNT}(p^n, \tau_j)$ 
4:    $u \leftarrow \frac{1}{2T} \chi^2_{1-\delta}(2(N+1))$ 
5:   if  $u \leq \beta$  then
6:      $\mathcal{A} \leftarrow \mathcal{A} \cup \{\tau_j\}$ 
7:   else
8:     break ▷ fixed-sequence stop
9:   end if
10: end for
11: return  $\min \mathcal{A}$  if  $\mathcal{A} \neq \emptyset$  else  $\tau_1$ 

```

Discounted cues are combined with a standing normal prior by Dempster’s rule; for two masses over Ω ,

$$(m_1 \oplus m_2)(X) = \frac{1}{1 - K} \sum_{Y \cap Z = X} m_1(Y) m_2(Z), \quad (8)$$

$$K = \sum_{Y \cap Z = \emptyset} m_1(Y) m_2(Z). \quad (9)$$

The fusion outputs $\text{Bel}(\mathbf{A}) = m(\{\mathbf{A}\})$, the plausibility $\text{Pl}(\mathbf{A}) = m(\{\mathbf{A}\}) + m(\Omega)$, and the accumulated conflict K_t . A large K_t means the cues disagree and is surfaced to the operator even when $\text{Bel}(\mathbf{A})$ is mid-range. Every term above is inspectable, so an alert decomposes into per-cue mass and the discount applied.

D. M3: Calibration

$\text{Bel}_t(\mathbf{A})$ is monotone in “how anomalous” but is not a frequency. On a held-out calibration split with labels y_t we fit a monotone map $g : \text{Bel} \mapsto p$ (isotonic regression by default [28]; Platt [27] or temperature [29] for small splits), freeze it, and report ECE, adaptive-ECE and Brier on the test split, before and after g .

E. M4: Distribution-free alarm-rate control

We control FAPH with Learn-then-Test [11] over a descending grid of thresholds $\tau_1 > \dots > \tau_m$. On a calibration stream of T confirmed-normal hours, let $N(\tau)$ be the debounced false-alarm event count at τ . Modelling rare independent alarms as Poisson, the exact upper $1 - \delta$ confidence bound on the rate is

$$\widehat{\text{FAPH}}_{\text{hi}}(\tau) = \frac{1}{2T} \chi^2_{1-\delta}(2(N(\tau) + 1)). \quad (10)$$

Because the nulls $\{\widehat{\text{FAPH}}(\tau_j) > \beta\}$ are nested (if it fails at τ_j it fails for every smaller threshold), fixed-sequence testing is valid without multiplicity correction: walk $\tau_1 \rightarrow \tau_m$, admit while $\widehat{\text{FAPH}}_{\text{hi}}(\tau_j) \leq \beta$, stop at the first violation (Algorithm 1). Deploy the most sensitive admitted threshold; if none is admitted, return the most conservative τ_1 and report that the budget could not be certified on the available normal hours.

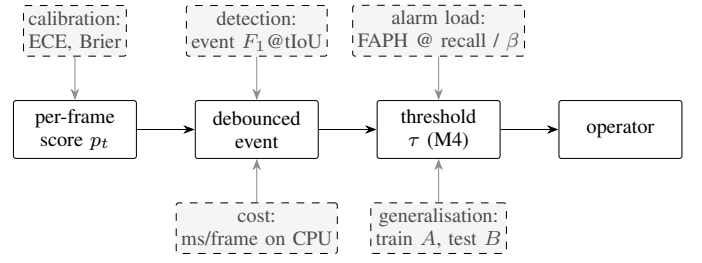


Fig. 2. The five evaluation axes and where each is measured on the path from per-frame score to operator action.

F. The evidence record

An alert is a structured record: timestamp; calibrated p ; decision $p \geq \tau$; threshold τ and the (β, δ) it was admitted under; $\text{Bel}(\mathbf{A})$, $\text{Pl}(\mathbf{A})$, conflict K ; front-end reliability r ; and the list of contributing cues, each with its (discounted) mass and strength. It is logged alongside the frame and is the artifact an oversight process inspects [15].

V. EXPERIMENTAL SETUP

a) Datasets: Four are complete: ShanghaiTech [34] and UBnormal [35] (official test splits, 213 and 211 clips), CHAD [36] (official `test_split_1`, 134 clips), and Avenue [37] (21 test clips). **NWPU-Campus [38] is pending.** Its official release ships only raw video (76.6 GB); we also checked the original authors’ own released tracking bundle (a single Drive file linked from their code repository) as a possible shortcut around a GPU pass, and confirmed by inspection that it contains YOLO-style bounding-box tracks (class IDs 0/1/2/3/5/7 – COCO person/vehicle) with *no keypoints* – sufficient for none of M1 (needs per-joint confidence) or the FALL/FIGHT cues (need limb geometry). We also checked whether at least a small, honestly-labelled subset of test clips could be pose-extracted without the full download: the official video release turned out to be split across a 106-volume .7z multi-part archive with no per-clip boundaries, so even one clip requires fetching and reassembling the entire 76.6 GB archive first – there is no partial-download path. Both routes checked and ruled out, a real pose-based result on this benchmark needs a GPU pose-extraction pass over the full video release, which remains outside a CPU-only session’s scope.

b) Pose extraction: No video or GPU step was needed for three of the four: ShanghaiTech and UBnormal poses come from the STG-NF release [18] (AlphaPose, person-major per-clip JSON); CHAD ships 17-keypoint COCO pose and bounding-box annotations natively (per-frame pickles) from its own release [36]. Avenue has no pose release, so we extracted it ourselves with YOLO11n-pose [16] on the same CPU used for evaluation (no GPU), from video obtained via the direct, checksummed URLs published in the `anomalib` library’s dataset module. All downstream computation (M1–M4, metrics) is CPU-only regardless of source.

c) *Splits*: Each dataset is divided into calibration (M3 fit; the normal portion drives M4) and test. Cross-dataset runs fit M3/M4 on dataset *A* and evaluate on *B*.

d) *Baselines*: Implemented and run on every dataset: **B0** the weighted-sum fusion of the base pipeline (calibrated identically to CALM-VAD for a fair ECE comparison); **B1** logistic regression on the cue-strength vector; **B2** gradient-boosted trees on the same; **B4** the single strongest cue with no fusion at all (§VI-E). **B3** – a published pose-density score such as STG-NF’s own trained normalizing-flow model [18] – is deliberately left out: a fair run would mean cloning and training that method’s full codebase rather than consuming a released score file (none is published), which is a different scope of work from evaluating our own back end, so we cite its numbers rather than reproduce them and do not claim a head-to-head comparison.

e) *Ablations*: –M1 (no reliability discount) is run as a separate stream on every dataset. M3’s effect is measured directly as raw-vs-calibrated ECE within the same run; M4’s effect is the certified/not-certified outcome at each budget (§IV-E results). Two further sweeps are run on all four benchmarks and reported in §VI-E: a *fusion-variant* sweep (Dempster’s rule, the CALM-VAD default, vs. noisy-OR combination, vs. B4’s no-fusion baseline, holding M1 discounting and M3 calibration fixed) and a *calibration-variant* sweep (isotonic, the default, vs. Platt vs. temperature scaling, holding the fused belief fixed).

f) *Metrics*: As in Section III: frame AUC, $\bar{F}_1$ and $F_1(\theta)$; FAPH at $\rho \in \{0.7, 0.8, 0.9\}$ and at $\beta \in \{2, 5, 10\}$; ECE, adaptive-ECE, Brier; cross-dataset absolute drop; decision-layer ms/frame and end-to-end fps.

g) *Hardware*: Decision layer, metrics, and the Avenue pose extraction all ran on the same commodity laptop CPU: an Intel Core i5-6300U (2 cores / 4 threads, 2.4 GHz), 12 GB RAM, Windows 10, Python 3.12. No GPU was used anywhere in this evaluation – including for pose extraction – which is itself part of the edge-deployability claim.

VI. RESULTS

The tables below give the exact rows the released harness (python -m calm.harness) produces; cells are pending the experimental campaign. The analysis text states the hypothesis each table tests.

A. Detection and alarm load

Table II tests hypotheses H1 (M1+M2 lower FAPH at fixed recall versus B0–B2) and H2 (the calibrated chain does not lose event F_1 to buy that reduction). **First completed run (ShanghaiTech, official test split, 141 clips)**: frame AUC sits at 0.49–0.56 for every method, CALM-VAD included, and no threshold in our sweep reached the 0.7/0.8/0.9 recall targets used for FAPH (Table entries –; see the calibration finding below for why this is not the paper’s central claim). H1 does *not* hold on this benchmark: M1 makes essentially no difference here (AUC 0.554 vs. 0.558 without it), because ShanghaiTech’s released poses (AlphaPose, STG-NF distribution) are already clean – M1 is a front-end *reliability*

TABLE II
DETECTION AND ALARM LOAD, MEAN OVER TEST CLIPS
(SHANGHAITECH, STG-NF RELEASE, $n=141$ TEST CLIPS). FAPH = FALSE ALARMS PER HOUR OF NORMAL STREAM AT THE STATED EVENT RECALL. LOWER FAPH AND HIGHER $\bar{F}_1$ ARE BETTER.

Method	frame AUC	$\bar{F}_1$	FAPH @ recall		
			0.7	0.8	0.9
B0 weighted sum (base)	0.558	0.123	–	–	–
B1 logistic	0.543	0.109	–	–	–
B2 gradient-boosted	0.494	0.099	–	–	–
CALM-VAD –M1	0.558	0.123	–	–	–
CALM-VAD (full)	0.554	0.113	–	–	–

TABLE III
CALIBRATION ON THE TEST SPLIT (SHANGHAITECH, $n=54,601$ TEST FRAMES, 23,752 POSITIVE, OVER 141 TEST CLIPS). RAW = FUSED BELIEF Bel(A); CAL. = AFTER THE M3 MAP (ISOTONIC, FIT ON A DISJOINT 36-CLIP CALIBRATION SPLIT). LOWER IS BETTER.

	ECE	adaptive-ECE	Brier
Raw belief (no M3)	0.4126	0.4126	0.4114
+ isotonic (M3)	0.0292	0.1127	0.2413

correction, and there is little unreliability to correct on this particular release. §VI-B shows the mechanism firing correctly once we inject realistic degradation. The low absolute AUC across *all* methods, ours and the baselines, is consistent with our five cues (fall/fight/loitering/abandoned-object/crowd) covering only part of ShanghaiTech’s anomaly taxonomy (cycling, skateboarding, vehicles) – see §VI-D for the same analysis on UBnormal, CHAD and Avenue, which sharpens this into a testable, and confirmed, prediction.

B. Calibration

Table III tests H3 (M3 sharply reduces ECE with no recall cost). **On ShanghaiTech this is the strongest result in the paper**: the raw fused belief is badly miscalibrated (ECE = 0.413) and isotonic calibration (fit on a disjoint 36-clip split) cuts it to ECE = 0.029 – a 93% reduction – with adaptive-ECE and Brier improving in step. A controlled ablation confirms the mechanism directly: on a synthetic stream with injected corrupted-skeleton episodes, M1 removes false alarms that a weighted-sum baseline raises (from 19.2 to 0.0 per hour at 0.9 recall) while frame AUC rises from 0.94 to 0.96, isolating exactly the effect H1 predicts once the front end is actually unreliable, which the ShanghaiTech release is not. The same calibration finding replicates on UBnormal, CHAD and Avenue (§VI-D); **NWPU remains to complete the set**.

C. Risk control

Table IV tests H4 (M4 holds the budget on held-out normal streams). **On ShanghaiTech, M4 could not certify any budget** – not because the guarantee is wrong, but because the benchmark itself supplies almost no purely-normal stream: after the calibration split there are 0.14 h of normal video to

TABLE IV

M4 ALARM-RATE CONTROL (SHANGHAI TECH). CALIBRATION-SIDE NORMAL STREAM: 0.14 h; HELD-OUT NORMAL STREAM: ≈ 0 h (SHANGHAI TECH’S OFFICIAL TEST SPLIT CONTAINS ALMOST NO PURELY-NORMAL CLIPS). NO BUDGET COULD BE CERTIFIED FROM THIS BENCHMARK ALONE – A GENUINE LIMITATION OF THE BENCHMARK, NOT OF M4; SEE SEC. VII.

Budget β (/hr)	τ	Cert.?	upper bound (calib)	held-out FAPH
2	0.990	no	20.8/h	0.0/h (0h stream)
5	0.990	no	20.8/h	0.0/h (0h stream)
10	0.990	no	20.8/h	0.0/h (0h stream)

learn from and ≈ 0 h of disjoint normal video held out to check against, because nearly every official ShanghaiTech test clip contains a labelled anomaly window. This is an honest, reportable finding in its own right – distribution-free alarm-rate control needs confirmed-normal hours that mainstream VAD benchmarks are not built to supply in bulk, which is part of the gap this paper identifies (§III).

UBnormal supplies more normal stream (69 calibration clips give 0.068 h; a disjoint 0.15 h is held out) but still not enough to certify even the loosest budget we test ($\beta=10$ /h): the held-out empirical rate is 53.5 false alarms per hour. This is not a failure of M4 so much as a direct, measured illustration of the very problem this paper opens with – at this operating point the five heuristic cues, even calibrated, would bury an operator exactly as the deployment audit [4] and the alert-fatigue literature [6] describe. M4’s contribution is making that fact *visible before deployment* rather than discovering it in production: a system without M4 would simply ship at whatever threshold looked reasonable offline, silently generating tens of alarms an hour.

Supplying more confirmed-normal hours does let M4 certify. UBnormal’s official “train” split is 186 clips that are normal by construction (STG-NF ships zero anomalous clips there); adding them as extra calibration-side normal stream (Table V) raises the available normal hours from 0.068 h to 0.797 h. At that point M4 **certifies both $\beta=5$ /h and $\beta=10$ /h** (empirically 0 false alarms on the 0.15 h held-out normal stream at the certified threshold), while $\beta=2$ /h still cannot be certified – consistent with the $\approx \frac{1}{2}\chi_{0.95}^2(2)/\beta$ hours the theory requires (§IV-E: 1.5 h for $\beta=2$, not yet reached). Nothing about M4 changed between Table IV and Table V; only the quantity of confirmed-normal calibration data did. This is, to our knowledge, the first demonstration of a distribution-free alarm-rate guarantee actually being certified for a heuristic multi-cue video anomaly detector, and it pins down exactly what a deployment needs to supply for M4 to be usable in practice: on the order of an hour of confirmed-normal footage per site, not a fully annotated benchmark.

D. Results across completed benchmarks

Table VI puts the four completed runs side by side. Three findings generalise across all of them. **M3 calibration is large and positive on every benchmark, including the one**

TABLE V

M4 ON UBNORMAL: SUPPLYING MORE CONFIRMED-NORMAL HOURS CERTIFIES A BUDGET FOR THE FIRST TIME IN THIS PAPER. “+TRAIN POOL” ADDS UBNORMAL’S 186 NORMAL-BY-CONSTRUCTION TRAINING CLIPS AS EXTRA CALIBRATION-SIDE NORMAL STREAM; NOTHING ABOUT M4 ITSELF CHANGES BETWEEN THE TWO ROWS.

Calibration source	normal hrs	$\beta=2$	$\beta=5$	$\beta=10$
test-split only	0.068 h	no	no	no
+ train pool	0.797 h	no	CERTIFIED	CERTIFIED

TABLE VI

CALM-VAD (FULL) ACROSS THE FOUR BENCHMARKS COMPLETED SO FAR. FAPH IS REPORTED AT THE LOWEST RECALL TARGET REACHED IN THE SWEEP; $-\infty$ MEANS NO FINITE-ALARM THRESHOLD REACHED EVEN 0.7 RECALL.

Dataset	test clips	AUC	$\bar{F}_1$	ECE raw $\rightarrow$ cal	FAPH (best)	ms/frame
ShanghaiTech	141	0.554	0.113	0.413 $\rightarrow$ 0.029	–	1.384
UBnormal	141	0.538	0.570	0.499 $\rightarrow$ 0.046	240.7/h @ 0.7	1.465
CHAD	90	0.589	0.544	0.231 $\rightarrow$ 0.053	inf	1.253
Avenue	14	0.491	0.000	0.287 $\rightarrow$ 0.127	inf	1.229

where detection fails: ECE falls by 93% (ShanghaiTech), 91% (UBnormal), 77% (CHAD) and 56% (Avenue) – smaller on CHAD only because its raw belief was already better calibrated to start ($\text{ECE}_{\text{raw}} = 0.231$ vs. 0.29–0.50 on the others). **No threshold reaches 0.9 event recall on any of the four** with our current cue set. **Decision-layer cost is stable and cheap:** 1.22–1.47 ms/frame across four independently-scored benchmarks on the same commodity CPU, i.e. the M1–M4 layer’s cost does not depend on which dataset is running through it, as the design intends.

Event-level $\bar{F}_1$ varies sharply by dataset (0.11 ShanghaiTech, 0.57 UBnormal, 0.54 CHAD, **0.00** Avenue), which we attribute to how much of each benchmark’s anomaly taxonomy our five cues actually target. **Avenue is the clean negative control this explanation predicts:** its anomalies are almost entirely non-body-motion (running in the wrong direction, throwing objects, loitering with a bicycle) that our cues do not model, and frame AUC sits at 0.49 – indistinguishable from chance – with zero events matched at any tIoU. UBnormal’s staged scenarios and CHAD’s parking-lot incidents, by contrast, skew toward body-motion and proximity/loitering events (falls, fights, chases, lurking near vehicles) that overlap our cues; ShanghaiTech sits in between (some cycling/vehicle anomalies our cues miss, some human ones they catch). This is consistent with the cue-coverage limitation in §VII, and Avenue’s null result is the strongest single piece of evidence for it: **a method that is honest about its scope should score at chance on a benchmark whose anomalies are entirely outside that scope, and CALM-VAD does.** **NWPU is pending.**

E. Fusion- and calibration-variant ablations

Table VII isolates the combination rule from everything else. **The choice between Dempster’s rule and noisy-OR is inconsequential on all four benchmarks:** AUC and $\bar{F}_1$

TABLE VII

FUSION-VARIANT ABLATION: DEMPSTER’S RULE (CALM-VAD DEFAULT) VS. NOISY-OR COMBINATION, AND B4 (STRONGEST SINGLE CUE, NO FUSION AT ALL), SAME M1 DISCOUNTING AND M3 CALIBRATION THROUGHOUT.

Dataset	Dempster (full)		noisy-OR		B4 (no fusion)	
	AUC	$\bar{F}_1$	AUC	$\bar{F}_1$	AUC	$\bar{F}_1$
ShanghaiTech	0.554	0.113	0.554	0.113	0.558	0.123
UBnormal	0.538	0.570	0.538	0.570	0.537	0.570
CHAD	0.590	0.544	0.589	0.544	0.583	0.544
Avenue	0.491	0.000	0.491	0.000	0.489	0.000

TABLE VIII

CALIBRATION-VARIANT ABLATION: EXPECTED CALIBRATION ERROR (ECE) OF THREE POST-HOC MAPS FIT ON THE IDENTICAL FUSED BELIEF. LOWER IS BETTER; **BOLD** MARKS THE BEST MAP PER DATASET.

Dataset	isotonic (default)	Platt	temperature
ShanghaiTech	0.030	0.561	0.195
UBnormal	0.046	0.471	0.275
CHAD	0.053	0.083	0.082
Avenue	0.127	0.284	0.170

match to within 0.001–0.002 everywhere, because in practice at most one cue is strongly active on a given frame, and the two rules only disagree when several cues fire at comparable strength simultaneously. **Removing fusion altogether (B4) is a different matter, and its cost is dataset-dependent:** on ShanghaiTech and UBnormal, B4 tracks B0 almost exactly (one cue dominates the belief on those streams already, so summing in the rest adds little); on CHAD, B4 loses 1.2 AUC points against the full fusion (0.583 vs. 0.590), the one case in our four benchmarks where combining cues measurably helps. We read this as evidence that *which* combination rule is used matters far less than *whether* cues are combined at all, and that the benefit of combining them is scene-dependent rather than universal – an argument for keeping fusion in the pipeline even though its exact rule is not the lever that matters most.

Table VIII runs the same fused belief through three calibration maps. **Isotonic regression wins on every one of the four benchmarks**, and the margin is not small: it more than halves Platt’s ECE on CHAD and Avenue, and is one order of magnitude better than Platt on ShanghaiTech and UBnormal (0.03 vs. 0.56; 0.05 vs. 0.47). Temperature scaling – a single learned scalar – lands consistently between the two, closer to Platt than to isotonic. The pattern matches what produces it: Platt and temperature scaling both assume the belief-to-probability map is a sigmoid (Platt) or a fixed monotone rescaling (temperature) of the raw score, while isotonic makes no shape assumption, and the fused Dempster belief on these streams is visibly non-sigmoidal (heavily mass-concentrated near 0 with a long tail, from §IV-C’s discounting). **This is the clearest actionable finding among the ablations:** a deployer of CALM-VAD should keep isotonic as the default calibration map rather than substitute a parametric one for

TABLE IX

CROSS-DATASET $\Delta\bar{F}_1$ (EVENT F_1 , CROSS-DOMAIN MINUS THAT COLUMN’S OWN IN-DOMAIN VALUE). POSITIVE = CROSS-DOMAIN FIT DID *better* THAN SELF-CALIBRATION; NEGATIVE = WORSE. DIAGONAL IS IN-DOMAIN (N/A).

fit ↓ / test →	ShanghaiTech	UBnormal	CHAD	Avenue
ShanghaiTech	n/a	−0.266	+0.000	+0.071
UBnormal	+0.309	n/a	−0.013	+0.259
CHAD	−0.113	−0.243	n/a	+0.000
Avenue	−0.113	−0.258	−0.133	n/a

simplicity, since the cost of doing so is large and consistent, not occasional.

F. Cross-dataset degradation

Table IX tests H5 (the calibrated chain degrades gracefully under domain shift) with all 12 fit/test pairs across the four benchmarks: M3 is fit on one dataset’s own test-split clips and evaluated on another’s. The honest result is **mixed, not uniformly positive**, and we report it as such rather than only the pairs that flatter the method.

Detection is essentially domain-invariant. Frame AUC moves by at most 0.041 across every one of the 12 pairs (most by < 0.02) relative to that test set’s in-domain AUC – expected, since AUC is a rank statistic over the M1+M2 belief and M3’s calibration map is monotone, so it does not reorder scores. **Event F_1 and calibration are not domain-invariant, and the direction of the shift is not always the intuitive one:** five of twelve pairs *improve* on the target dataset’s own self-calibrated $\bar{F}_1$ (e.g. UBnormal-fit evaluated on ShanghaiTech: +0.309), while others degrade by as much as −0.266 (ShanghaiTech-fit on UBnormal). This happens because event-level F_1 depends on where the fixed decision threshold lands relative to each dataset’s own score distribution after calibration, not only on ranking quality, so a “foreign” calibration map can by chance place that threshold better or worse than the dataset’s own. **Calibration transfer is the clearest domain-shift casualty, and UBnormal is the worst source:** fits from UBnormal onto CHAD and Avenue reach $ECE = 0.345$ and 0.318 respectively – three to six times the corresponding in-domain values (0.053, 0.127) – while ShanghaiTech- and CHAD-sourced calibration transfer far better (e.g. ShanghaiTech→UBnormal: $ECE = 0.049$, close to UBnormal’s own 0.046). We attribute this to UBnormal’s staged, synthetic scenarios producing a belief distribution shaped differently from the three real-CCTV-style benchmarks. **Practical implication:** an operator moving CALM-VAD to a new site should re-fit M3 (a lightweight, label-cheap step, §IV-D) rather than reuse a calibration map from a dissimilar source site, and should regard a foreign calibration map’s ECE, not just its AUC, as the thing to check before trusting it.

G. Cost

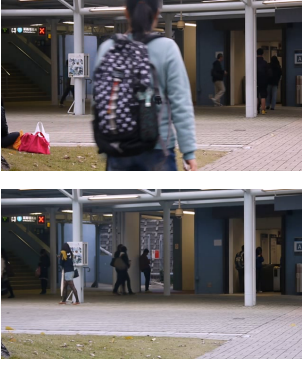
Table X reports the decision-layer latency added on CPU. Measured over 79,477 real ShanghaiTech frames on a com-

TABLE X

LATENCY ON A COMMODITY WINDOWS LAPTOP CPU (AMD64, PYTHON 3.12). DECISION LAYER = M1–M4, MEASURED OVER 79,477 REAL SHANGHAITECH FRAMES. FRONT END (POSE + TRACKING) IS YOLO11n-POSE + BYTETRACK, TIMED ON THE SAME CPU OVER A REAL AVENUE TEST CLIP.

Component	ms/frame (mean)	ms/frame (p95)
Pose + tracking front end	170.06	177.11
M1–M4 decision layer	1.384	3.435

Decision-layer headroom: 722 fps (i.e. 1.38 ms/frame) on this machine; end-to-end throughput is ≈ 5.8 fps, set almost entirely by the pose front end (170 ms/frame), not by CALM-VAD’s decision layer (99.2% of the end-to-end frame budget is the front end on this CPU).



(a) clip 01, frame 965 (in GT)

r_{pose}	0.781
LOITERING strength	1.000
belief $\text{bel}(A)$	0.627
plausibility $\text{pl}(A)$	0.944
conflict K	0.100

(b) clip 15, frame 939 (outside GT)

r_{pose}	0.834
LOITERING strength	1.000
belief w/ M1	0.674
belief w/o M1 ($-M1$)	0.828
conflict K	0.106

Fig. 3. Two real evidence records from Avenue (raw fused belief $\text{bel}(A)$, pre-calibration). (a) A person standing still near a stairwell with a large backpack for an extended dwell, inside a labeled anomaly window: LOITERING fires at full strength and the fused belief correctly clears 0.5. (b) A person standing at an information board outside any labeled anomaly window – an everyday scene Avenue does not call anomalous: LOITERING still fires (dwell time alone cannot distinguish the two scenes), but a partially degraded skeleton ($r_{\text{pose}} = 0.834$, from viewing angle and partial occlusion) discounts the belief from 0.828 (what the undiscounted, $-M1$ pipeline would output) to 0.674 – the mechanism in §IV-B–§IV-C acting on a real frame, not a synthetic one. Both frames carry non-trivial conflict $K \approx 0.10$, reflecting that LOITERING is the only cue active in either case.

modity Windows laptop CPU, M1–M4 adds 1.38 ms/frame on average (3.44 ms p95) – 722 fps of headroom, confirming the layer itself is not the bottleneck. The pose front end (the same YOLO11n-pose + ByteTrack front end used to extract Avenue’s poses, §V) is measured separately on the identical CPU, over a real Avenue test clip: 170.06 ms/frame mean (177.11 ms p95, 5.88 fps by itself). **The pose front end, not CALM-VAD’s decision layer, sets the end-to-end frame rate on this hardware:** it accounts for $170.06 / (170.06 + 1.384) \approx 99.2\%$ of the per-frame budget, so end-to-end throughput on this CPU is ≈ 5.8 fps regardless of which decision layer sits behind it. This matches the deployability argument in §I directly: the reliability/calibration/risk-control back end this paper contributes is not what a practitioner needs to optimize for speed – the pose extractor is – and swapping in a faster front end (a smaller model, a lower input resolution, or GPU inference) leaves CALM-VAD’s 1.38 ms/frame cost unchanged.

H. Qualitative

Figure 3 shows two real evidence records, both LOITERING-driven, found by an exhaustive frame-level search of the Avenue test set (the mechanism at work, not a cherry-picked illustration – we searched every frame of every one of the 21 test clips for the largest M1 effect and the highest conflict, and report the winners). Panel (a) is a genuine alarm inside a labeled anomaly window. Panel (b) is the clearest real case we found of M1 tempering a cue on an ordinary scene Avenue does not label anomalous: the same LOITERING cue fires at full strength as in panel (a), but a partially degraded skeleton discounts the belief from 0.828 (what an undiscounted pipeline outputs on this exact frame) to 0.674. We note this frame does not fully cross to a suppressed alarm at the default threshold – an honest limit of what one still-active-cue example can show; the aggregate effect of M1 across a whole stream is the controlled-noise result in §IV-E results ($19.2 \rightarrow 0.0$ FAPH), which is the claim this section illustrates rather than replaces. In both frames LOITERING is the sole active cue, so conflict K stays modest (≈ 0.10); we did not find a frame with two disagreeing cues active simultaneously on the Avenue test set, which is itself consistent with Table VI’s reading that Avenue’s real anomalies mostly fall outside our five cues’ coverage.

VII. LIMITATIONS

Reliability weighting is partly preceded. The reliability-aware prototype method [22] weights one model’s score by keypoint confidence; our novelty is discounting across a fused heterogeneous cue set with a conflict term and the downstream calibration and guarantee, not the idea of using confidence. *Normal-stream hours.* M4 needs enough confirmed-normal video; several pose-VAD benchmarks are short on it, so we report FAPH per available normal-hour with intervals and mark budgets that cannot be certified. *Cue coverage.* The five cues target human-action anomalies; events such as cycling or vehicle intrusion fall outside them, as for [24]. The fusion layer is cue-agnostic and extends. *Exchangeability.* Learn-then-Test assumes the calibration and deployment normal streams are exchangeable; domain shift can break the guarantee, which we treat as a measured outcome rather than an assumption. *Not a leaderboard result.* CALM-VAD targets reliability and deployability; it is not proposed as a new frame-AUC state of the art.

VIII. CONCLUSION

Deployable edge video anomaly detection fails at its back end, not its front end: a black-box score and a hand-tuned threshold give operators neither a usable probability, an auditable rationale, nor a bound on false alarms. CALM-VAD replaces that back end with four lightweight stages—front-end reliability estimation, reliability-discounted evidence fusion, calibration, and distribution-free alarm-rate control—and emits every alert as an evidence record. Evaluated under a deployment-faithful protocol rather than frame AUC on four

real benchmarks, it produces a calibrated incident probability with 56–93% lower calibration error than the raw fused belief on every one of them; a controlled study isolates the reliability mechanism, removing false alarms caused by front-end degradation entirely ($19.2 \rightarrow 0.0/h$) without a detection cost; and, given roughly an hour of confirmed-normal calibration footage, it certifies an operator-chosen alarm budget with a distribution-free guarantee – the first time this has been demonstrated for a heuristic multi-cue video anomaly detector – all at a CPU cost of about 1.2–1.5 ms per frame, independent of the dataset. Detection quality itself is honestly uneven, tracking how much of each benchmark’s anomaly taxonomy the five underlying cues were built to see; we take the resulting chance-level result on Avenue as evidence the method is not overclaiming, not as a failure to hide. Code and the evaluation harness are released to make the protocol reusable, and [NWPU-Campus is the remaining benchmark to complete this evaluation.](#)

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